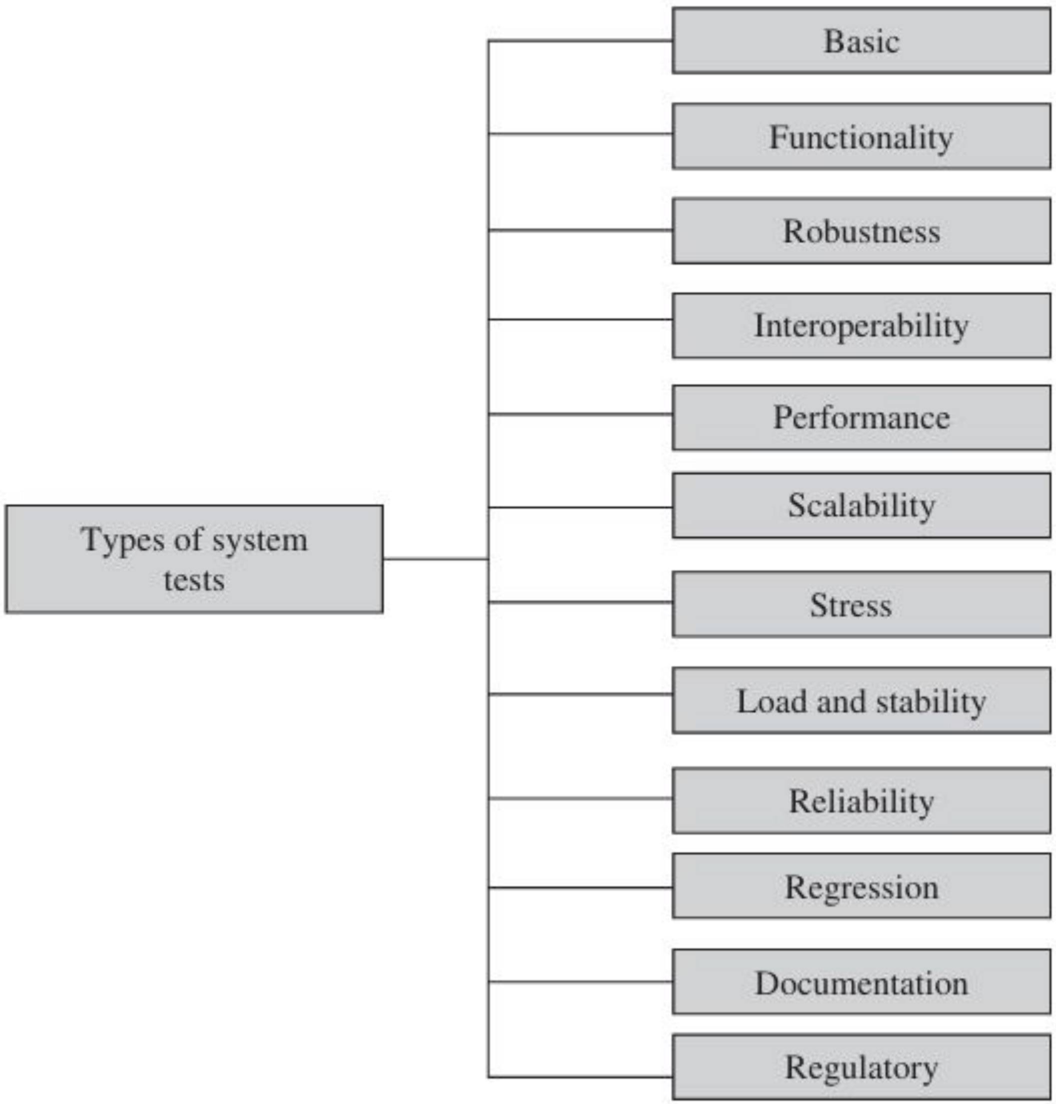
System Testing

System Testing

- Definition: The testing of a complete, integrated software product to evaluate the system's compliance with its specified requirements.
- Goal: To confirm that the fully assembled system works as designed, meets user needs, and satisfies non-functional criteria.
- Scope: Addresses defects arising from component interaction, data flow, external interfaces, and resource utilization.

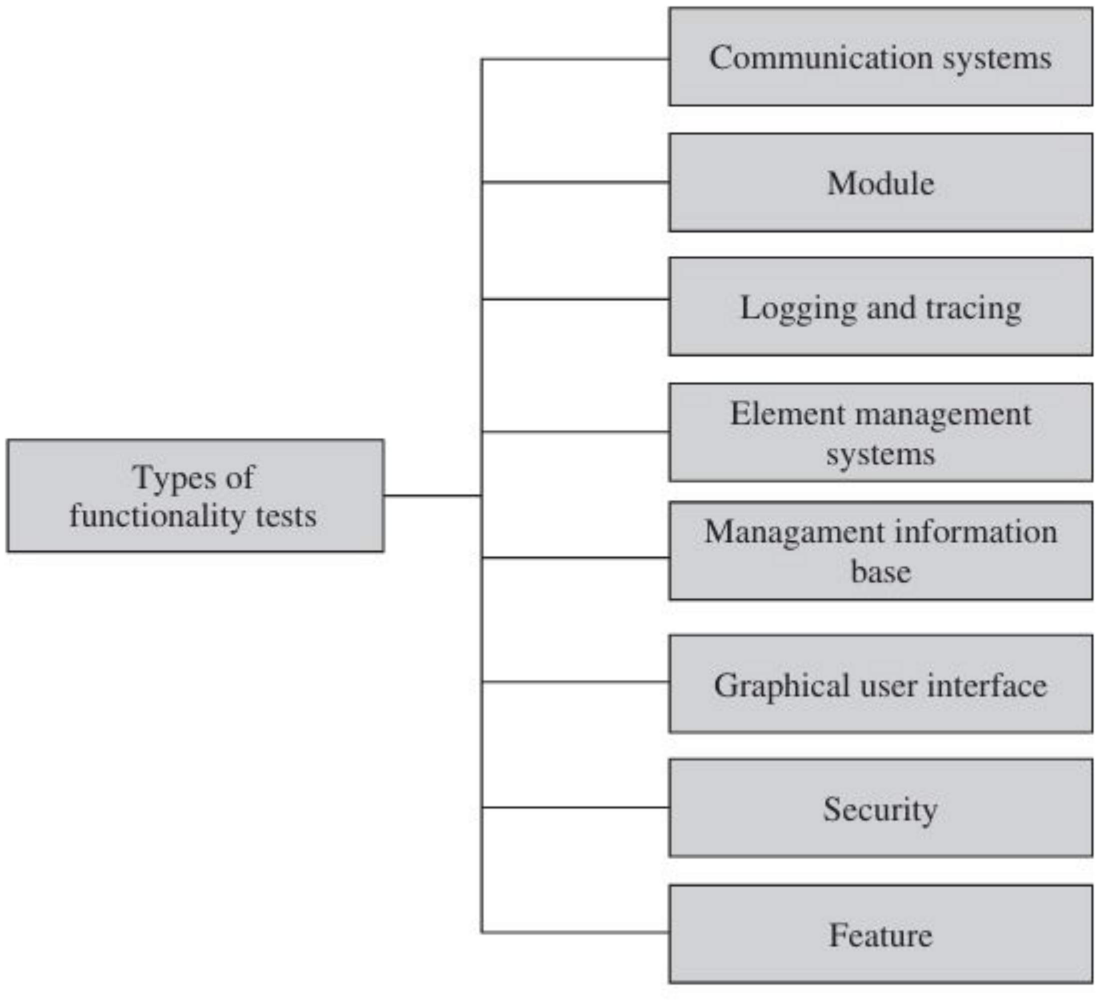
System Test Taxonomy



Basic Tests

- **Purpose:** To verify that the system can be **installed, configured, and brought to a basic operational state**. Serves as a gateway for full system testing.
- Boot/Power-On Tests:
- **Upgrade/Downgrade Tests:**
- **Light Emitting Diode Tests:**
- **Diagnostic Tests**
  - **Power-On Selft-Test (POST)**
  - **Bit Error Test (BERT)**
- **Command Line Interface Test**

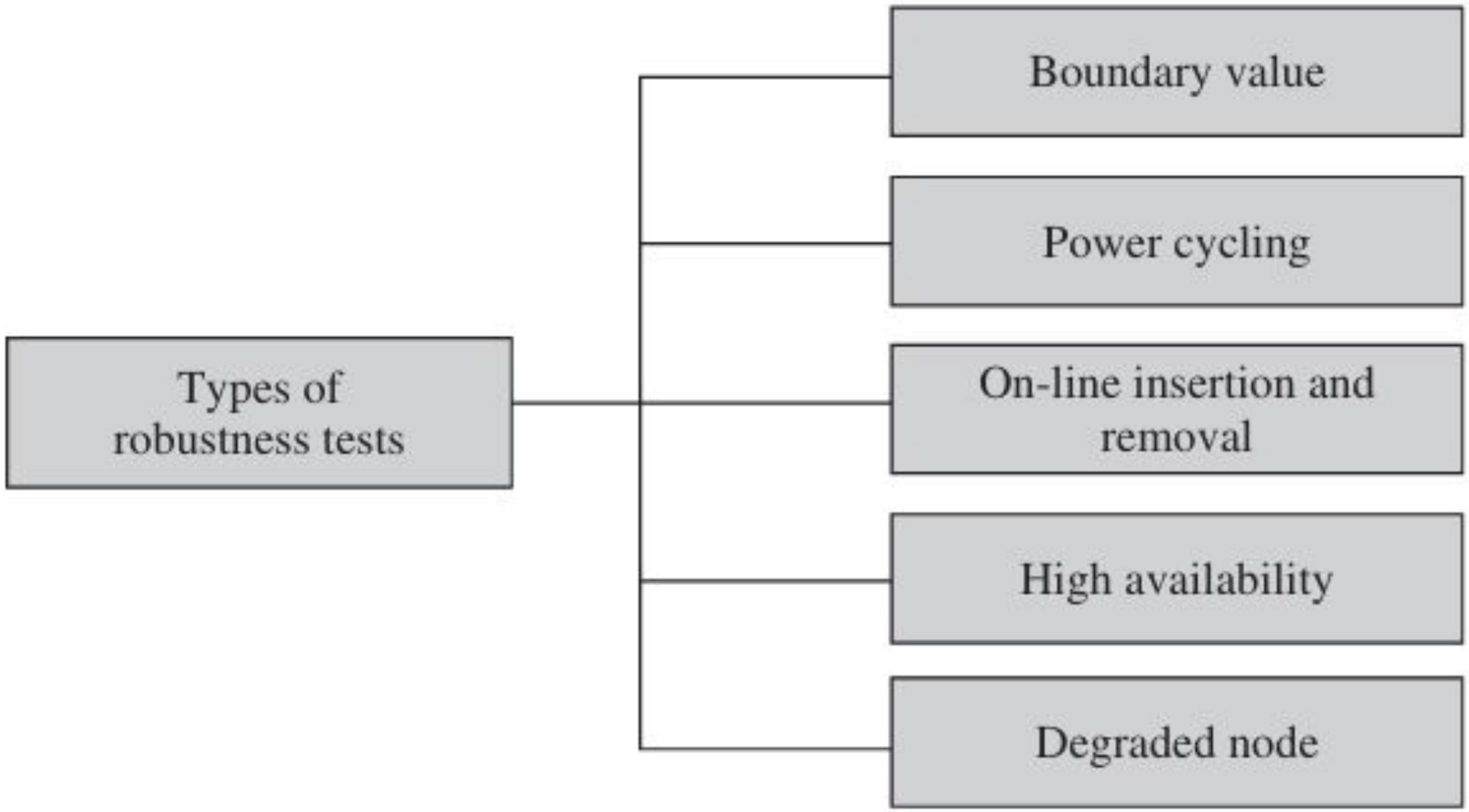
Functionality Tests



ROBUSTNESSTESTS

- Robustness means how sensitive a system is to erroneous input and changes in its operational environment.
- Tests in this category are designed to verify how gracefully the system behaves in error situations and in a changed operational environment.
- The purpose is to deliberately break the system, not as an end in itself, but as a means to find error.
- It is difficult to test for every combination of different operational states of the system or undesirable behavior of the environment.

ROBUSTNESSTESTS



Interoperability Tests

- In this category, tests are designed to verify the ability of the system to interoperate with third-party products.
- An interoperability test typically combines different net work elements in one test environment to ensure that they work together.
- In other words, tests are designed to ensure that the software can be connected with other systems and operated.
- In many cases, during interoperability tests, users may require the hardware devices to be interchangeable, removable, or reconfigurable. Often, a system will have a set of commands or menus that allow users to make the configuration changes.
- The reconfiguration activities during interoperability tests are known as configuration testing.
- Another kind of interoperability test is called a(backward) compatibility test.
- Compatibility tests verify that the system works the same way across different platforms, operating systems, and database management systems.
- Backward compatibility tests verify that the current software build f lawlessly works with older version of platforms.

Performance Tests

- Performance tests are designed to determine the performance of the actual system compared to the expected one.
- The performance metrics needed to be measured vary from application to application.
- A trans action in an on-line system requires a response of less than 1 second 90% of the time.
  - then one needs to capture (i) end-to-end response time (as seen by external user), (ii) CPU time, (iii) network connection time, (iv) database access time, (v) network connection time, and (vi) waiting time.

Scalability Tests

- All man-made artifacts have engineering limits.
- In this group, tests are designed to verify that the system can scale up to its engineering limits.
- A system which works acceptably at one level of demand may not scale up to another level.
- Scaling tests are conducted to ensure that the system response time remains the same or increases by a small amount as the number of users are increased.
- Systems may scale until they reach one or more engineering limits.

Stress Tests

- The goal of stress testing is to evaluate and determine the behavior of a software component while the offered load is in excess of its designed capacity.
- The system is deliberately stressed by pushing it to and beyond its specified limits.
- Stress tests include deliberate contention for scarce resources and testing for incompatibilities.
- It ensures that the system can perform acceptably under worst-case conditions under an expected peak load.
- If the limit is exceeded and the system does fail, then the recovery mechanism should be invoked.

Load and Stability Tests

- Load and stability tests are designed to ensure that the system remains stable for a long period of time under full load.
- A system might function flawlessly when tested by a few careful testers who exercise it in the intended manner.
- However, when a large number of users are introduced with incompatible systems and applications that run for months without restarting, a number of problems are likely to occur:
  - (i) the system slows down, (ii) the system encounters functionality problems, (iii) the system silently fails over, and (iv) the system crashes altogether.

RELIABILITY TESTS

- Reliability tests are designed to measure the ability of the system to remain operational for long periods of time.
- The reliability of a system is typically expressed in terms of mean time to failure (MTTF).
- As we test the software and move through the system testing phase, we observe failures and try to remove the defects and continue testing.
- As this progresses, we record the time durations between successive failures.

Regression Tests

- In this category, new tests are not designed. Instead, test cases are selected from the existing pool and executed to ensure that nothing is broken in the new version of the software.
- The main idea in regression testing is to verify that no defect has been introduced into the unchanged portion of a system due to changes made elsewhere in the system.
- During system testing, many defects are revealed and the code is modified to fix those defects.

Documentation Tests

- Documentation testing means verifying the technical accuracy and readability of the user manuals, including the tutorials and the on-line help.
- Documentation testing is performed at three levels as explained in the following:
- Read Test: In this test a documentation is reviewed for clarity, organization, f low, and accuracy without executing the documented instructions on the system.
- Hands-On Test: The on-line help is exercised and the error messages verified to evaluate their accuracy and usefulness.
- Functional Test: The instructions embodied in the documentation are followed to verify that the system works as it has been documented.

Regulatory Tests

- In this category, the final system is shipped to the regulatory bodies in those countries where the product is expected to be marketed.
- The idea is to obtain compliance marks on the product from those bodies.
- HIPPA – Oxley-ACT etc Compliance